

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Digital Systems

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section “A”

Very Short Answer Questions

Attempt all the questions. [$10 \times 2 = 20$]

1. Briefly explain any two major features of a digital system.
2. Explain parity bit with example.
3. Draw the logic symbol and write the truth table for a two-input XNOR gate.
4. Convert (1011) Gray code to its equivalent BCD.
5. Define decoder.
6. Differentiate between a latch and a flip-flop.
7. Define Combinational circuit with block diagram.
8. Define toggling mode of JK Flip Flop.
9. Define minterm and maxterm.
10. Draw the block diagram of a 2-to-4 line decoder and explain how it works.

Section “B”

Descriptive Answer Questions

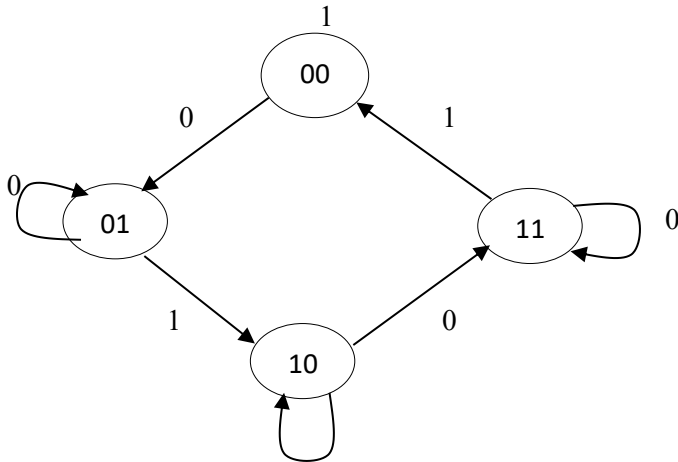
Attempt **any five** questions. [$5 \times 10 = 50$]

11. Simplify the given function:
 $F = \pi(0, 2, 5, 7, 9)$ and $\pi d(1, 4, 8, 9, 12)$
Using k-map and draw the logic diagram using
 - a. Basic gates only
 - b. NOR gates only
12. What is multiplexer? Implement 8 x 1 MUX using 2 x 1 MUX.
13. Differentiate between half adder and full adder. Design a Full Subtractors circuit. Your answer must include the truth table, logical expressions for Difference (D) and Borrow (Bout), and the final logic circuit diagram.
14. Define code converter. Design a combination circuit which takes 4-bit BCD input and displays equivalent excess-3 code of input. [2+8]
15. Compare and contrast synchronous and asynchronous counters, highlighting at least two advantages and two disadvantages of each. Design a MOD-5 asynchronous (ripple) counter using J-K flip-flops.
16. Design a 4-bit arithmetic circuit and explain its operation.

Long Answer Questions

Attempt **any two** questions. [$2 \times 15 = 30$]

17. Design the clocked sequential circuit using JK flip-flops with the help of given state diagram.



18. Design MOD-12 synchronous up counter with state diagram and timing diagram.
19. Explain the block diagram of 4-bit ALU. Explain the process of designing arithmetic unit of 4-bit ALU using parallel adder and also show necessary expressions and logic diagrams.